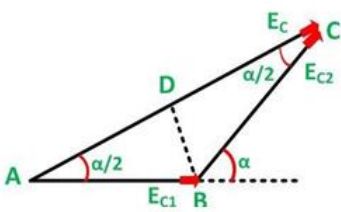
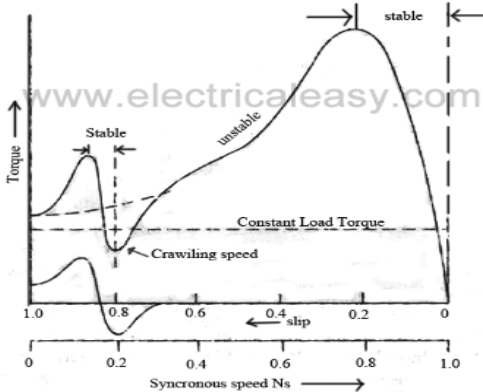


Sahrdaya College of Engineering and Technology, Kodakara																				
Answer Key																				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY																				
FIFTH SEMESTER B.TECH DEGREE EXAMINATION, JANUARY 2023																				
Course Code: EET307																				
Course Name: SYNCHRONOUS & INDUCTION MACHINES																				
PART A																				
	Answer all questions. Each question carries 3 marks																			
Q1	Compare salient pole and cylindrical rotor type of alternators.	3 marks																		
Ans	<table><thead><tr><th>Salient pole rotor</th><th>Cylindrical Rotor</th></tr></thead><tbody><tr><td>Poles are projecting out from the surface</td><td>Unslotted portion act as poles and are non projecting</td></tr><tr><td>Non uniform air gap</td><td>Uniform air gap</td></tr><tr><td>Large diameter and small axial length</td><td>Small diameter and large axial length</td></tr><tr><td>Windage loss as well as noise is more</td><td>Windage loss as well as noise is less as compared to salient pole rotors</td></tr><tr><td>Preferred for low speed alternators</td><td>Preferred for high speed alternators – turbo alternators</td></tr><tr><td>Prime mover used are water turbines and IC engines</td><td>Prime mover used are steam turbines and electric motors</td></tr><tr><td>For same size rating is smaller than cylindrical rotor</td><td>For same size rating is higher than salient type</td></tr><tr><td>Mostly used in hydro power plants</td><td>Used in nuclear, gas and thermal power plants</td></tr></tbody></table>	Salient pole rotor	Cylindrical Rotor	Poles are projecting out from the surface	Unslotted portion act as poles and are non projecting	Non uniform air gap	Uniform air gap	Large diameter and small axial length	Small diameter and large axial length	Windage loss as well as noise is more	Windage loss as well as noise is less as compared to salient pole rotors	Preferred for low speed alternators	Preferred for high speed alternators – turbo alternators	Prime mover used are water turbines and IC engines	Prime mover used are steam turbines and electric motors	For same size rating is smaller than cylindrical rotor	For same size rating is higher than salient type	Mostly used in hydro power plants	Used in nuclear, gas and thermal power plants	Any 3 points
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Q2	Briefly explain the term pitch factor in an alternator and derive an expression for it.	3 marks																		
Ans	- It is a measure of reduction in the induced emf due to short chording coils/ short pitched coils. - It is the ratio of emf induced in the short pitched coils to that of full pitched coils. $E_{C1} = E_{C2} = E$ $AD = DC = E \cos(\alpha / 2)$ $AC = 2E \cos(\alpha / 2)$ $K_p = \frac{2E \cos(\alpha / 2)}{2E} = \cos\left(\frac{\alpha}{2}\right)$ 																			

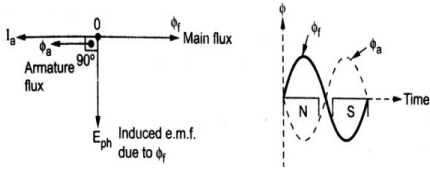
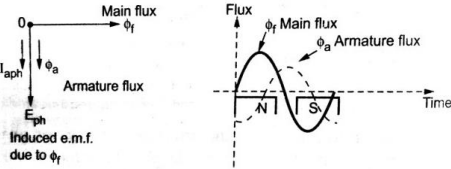
Q3	Draw the phasor diagram of a salient pole type alternator with lagging power factor load.	3 marks
Ans	* With lagging pf load	
Q4	X_d and X_q are different for salient pole alternators where as they are same for cylindrical rotor alternators. Justify.	3 marks
Ans	In a cylindrical rotor type machine, air gap is uniform so that the reluctance of the magnetic path is the same in all directions. Hence, effect of armature reaction can be accounted by one reactance X_s. X_s is constant for all directions of armature flux relative to rotor. However, in a salient pole type machine, the radial length of the air gap varies so that the reluctance of the magnetic circuit along polar axis (direct axis) is much less than the reluctance along the interpolar axis (quadrature axis). Because of the lower reluctance along the polar axis, more flux is produced along d-axis than along q-axis. Hence, reactance due to armature reaction will be different along d-axis and q-axis.	
Q5	Synchronous motor is not self starting. Why?	3 marks
Ans	• Whenever three phase supply is given to the 3-phase windings, a rotating magnetic field will be created. • The speed of rotation will be equal to the synchronous speed. • Rotating magnetic field can be assumed to be equivalent to the physical rotation of two stator poles N_1 and S_1. • Consider an instant as shown in figure (a). • At this instant rotor is stationary. Since similar poles repel each other, rotor will experience a torque in anticlockwise direction. • Now the stator poles are rotating very fast, that is at synchronous speed, the stator poles changes their positions at the same speed. • Due to large inertia of rotor, before rotor hardly rotates in the direction of anticlockwise torque, stator poles change their positions as shown in figure (b).	

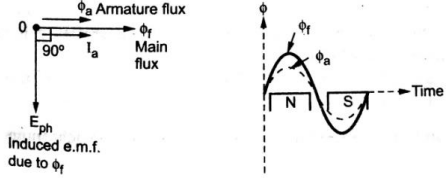
	- But before this happens, stator poles again change their positions and this process continues. - As a result average torque exerted on the rotor is zero. And hence the synchronous motor is not self starting.	
	(a) Action of synchronous motor (b) Action of synchronous motor	
Q6	A 3 phase, 50 Hz induction motor, has a full load speed of 700 rpm and a no load speed of 740 rpm. (a) How many poles does the machine have? (b) Find the slip and rotor frequency at full load? (c) What is the speed of the rotor field at full load (i) with respect to rotor (ii) with respect to stator	3 marks
Ans	a) No load speed = 740 rpm The very next synchronous speed, $N_s = 750$ $N_s = 120f / P$ $750 = (120 \times 50) / P$ $P = (120 \times 50) / 750 = 8$ b) Slip, $s = (N_s - N) / N_s = (750 - 700) / 750 = 0.067$ Rotor frequency = $sf = 0.06 \times 50 = 3.33 \text{ Hz}$ c) Speed of rotor field with respect to rotor = $750 - 700 = 50 \text{ rpm}$ Speed of rotor field with respect to stator = 750 rpm	
Q7	Consider a 3 phase Induction motor with a normal torque speed characteristics. Neglecting the effects of stator resistance and leakage reactance, discuss the approximate effect on the characteristics if the applied voltage is kept constant, but the frequency is halved.	3 marks
Ans	$f_1 > f_2 > f_3$	

	Reducing the frequency while keeping the voltage constant will increase the slip of the motor, leading to an increase in torque production but a decrease in power factor.	
Q8	Briefly explain crawling in Induction motor	3 marks
Ans	3 phase induction motor exhibits a tendency to run at very slow speeds (as low as one-seventh of their synchronous speed). This phenomenon is called as crawling of an induction motor. - This action is due to the fact that, flux wave produced by a stator winding is not purely sine wave. Instead, it is a complex wave consisting a fundamental wave and odd harmonics like 5th, 7th etc. - The fundamental wave revolves synchronously at synchronous speed N_s whereas 5th, 7th harmonics may rotate in forward or backward direction at $N_s/5$, $N_s/7$ speeds respectively. - Hence, harmonic torques are also developed in addition with fundamental torque. - The rotating magnetic field produced by 5th harmonic component will rotate at a $N_s/5$ and in a direction opposite to that of fundamental component, whereas the rotating magnetic field produced by 7th harmonic component will rotate at a $N_s/7$ and in the same direction as that of fundamental component. - Thus, motors exhibits a tendency to run at a speed nearly equal to $1/7^{\text{th}}$ of synchronous speed. - This phenomenon is called Crawling 	
Q9	Explain the principle of operation of induction generator	3 marks
Ans	An induction generator is a type of AC generator that uses an induction machine, also known as an asynchronous generator, to convert mechanical power into electrical power. Unlike synchronous generators, which require a separate DC power source to establish a magnetic field, induction generators generate their own magnetic field using the induction principle.	

Ans	The induction principle states that when a conductor is placed in a changing magnetic field, a current is induced in the conductor. In an induction generator, the rotor is connected to a prime mover, such as a wind turbine or water turbine, and rotates at a speed slightly less than the synchronous speed of the machine. The stator is wound with a set of stationary coils and is connected to the electrical load. As the rotor rotates, the magnetic field produced by the rotor windings cuts across the stationary stator windings, inducing a current in the stator windings. The induced current in the stator windings produces a magnetic field that opposes the magnetic field produced by the rotor windings, resulting in an electromagnetic torque that drives the rotor. The speed of the rotor determines the frequency of the induced voltage, which is proportional to the synchronous speed of the machine and the number of poles. The induced voltage is then rectified and converted into a DC voltage using a rectifier circuit, which is then used to supply power to the electrical load.	
Q10	Enumerate few applications of single phase induction motor	3 marks
Ans	• Domestic appliances: Single-phase induction motors are used in various domestic appliances such as washing machines, air conditioners, refrigerators, ceiling fans, and vacuum cleaners. • Compressors: Single-phase induction motors are used in air compressors for applications such as filling tires, powering pneumatic tools, and inflating balloons. • Agriculture: Single-phase induction motors are used in agricultural applications such as irrigation pumps, fodder cutting machines, and grain mills. • Industrial equipment: Single-phase induction motors are used in industrial equipment such as conveyor belts, blowers, mixers, and grinders. • Office equipment: Single-phase induction motors are used in office equipment such as paper shredders, printers, and scanners. • Medical equipment: Single-phase induction motors are used in medical equipment such as dental drills, X-ray machines, and surgical instruments. • HVAC systems: Single-phase induction motors are used in HVAC (heating, ventilation, and air conditioning) systems for applications such as air handlers, exhaust fans, and condenser fans. • Automotive industry: Single-phase induction motors are used in automotive applications such as windshield wipers, electric mirrors, and power windows.	Any 3 applications (1 mark each)
PART B		
<i>Answer any one full question from each module. Each question carries 14 marks</i>		
Module 1		
Q11 (a)	Explain the causes of harmonics in an alternator? How it can be minimised?	6 marks

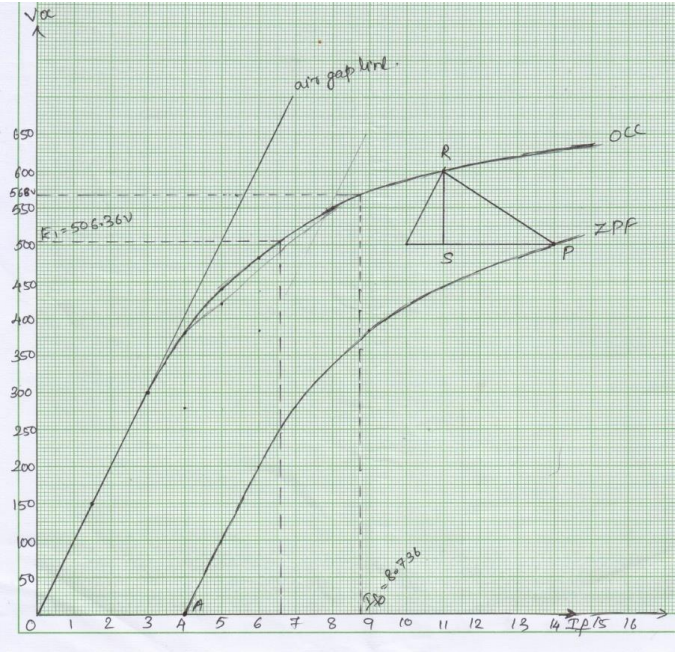
Ans	Causes of Harmonics 1) Overheating of electrical equipments 2) High voltages and circulating currents 3) Equipment malfunction due to distortion 4) False tripping of circuit breaker 5) Metering error and generator failure etc Methods to suppress or minimize harmonics 1) Distribution of armature windings : 2) Chording : 3) Skewing: 4) Large length of air gap	
Q11 (b)	A 3-phase, 4 pole, 50 Hz, synchronous generator has 48 slots in which double layer winding is housed. Each coil has 10 turns short pitched by an angle of 36° electrical. Flux/pole is 0.025 Wb (sinusoidally distributed). Then, for a 3phase, Y connection, find (i) the line to line induced emf (ii) the fifth harmonic component of line to line induced emf.	8 marks
Ans	(i) $EMF = 4.44 K_w \phi T_p$ $K_w = K_d \times K_p$ $K_p = \cos(\alpha / 2) = \cos(36 / 2) = 0.951$ $K_d = \frac{\sin\left(\frac{m\beta}{2}\right)}{m \sin\left(\frac{\beta}{2}\right)}$ $m = \text{slots} / \text{pole} / \text{phase} = 48 / 4 / 3 = 4$ $\beta = 180 / (S / P) = 180 / (48 / 4) = 15^\circ$ $K_d = \frac{\sin\left(\frac{4 \times 15}{2}\right)}{4 \sin\left(\frac{15}{2}\right)} = 0.9577$ $K_w = K_d \times K_p = 0.9107$ $T_p = \frac{48 \times 10}{2 \times 3} = 80$ $EMF / \text{phase} = 4.44 K_w \phi T_p = 4.44 \times 0.9107 \times 0.025 \times 50 \times 80 = 404.351V$ $EMF_{\text{line-line}} = \sqrt{3} \times 404.351 = 700.356V$	

	(ii) For 5th harmonic component, pitch factor $K_p = \cos(5\alpha / 2) = \cos(5 \times 36 / 2) = 0$ Hence, 5th harmonic voltage = 0	
Q12 (a)	Explain the effect of armature flux on main field flux when an alternator is operating at (i) lagging pf. (ii) Unity pf (iii) Leading pf.	10 marks
Ans	- When the load is connected to the alternator, the armature winding of the alternator carries current. - Every current carrying conductor produces its own flux. - Thus current carrying armature winding also produces its own flux. - So, there are two fluxes present in the air-gap of alternator: Flux produced by the field system (Main flux) and the flux produced by the current carrying armature windings (Armature flux). The effect of armature flux on the main flux is called armature reaction. - The effect of armature flux depends not only on the magnitude of the armature current but also on the power factor of the load connected to the alternator. (i) Lagging pf - Armature flux and main flux are exactly in opposite direction. - This causes reduction in the magnitude of main flux. - So induced emf and thus terminal voltage of alternator drops. - This effect of armature reaction under zero lagging power factor condition is called demagnetising effect of armature reaction.  (ii) UPF - Armature flux and main flux are 90 degree out of phase. - Thus main flux gets distorted. - This distorting effect of armature reaction under UPF condition is called cross magnetising effect of armature reaction. 	

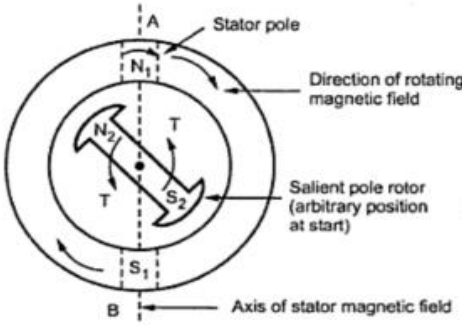
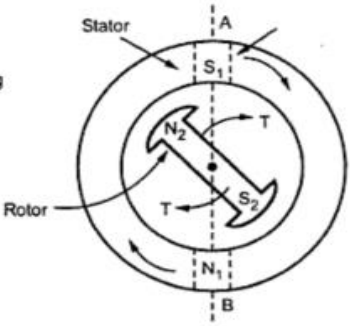
	(iii) Leading pf - Armature flux and main flux are in the same direction. - This causes addition in the magnitude of main flux. - So induced emf and thus terminal voltage of alternator increases. - This effect of armature reaction under zero leading power factor condition is called magnetising effect of armature reaction. 	
Q12 (b)	Derive the emf equation of an alternator.	4 marks
Ans	Let, ϕ : Flux per pole in Weber (Wb) Z_p : Total number of armature conductors / phase P : Number of poles N : Speed of rotation in rpm f : Frequency of generated EMF T_p : Armature turns per phase K_d : Distribution factor K_p : Pitch factor K_w : Winding factor Flux cut by one conductor in one revolution of rotor, $d\phi = P\phi$ Time taken to complete one revolution, $dt = 60/N$ sec Average EMF generated in one conductor, $\frac{d\phi}{dt} = \frac{P\phi}{60/N} = \frac{PN\phi}{60}$ Thus, average EMF induced per phase, $E_{ph} = \frac{PN\phi}{60} \times Z_p = 2f\phi \times Z_p$ $= 2f\phi \times 2 \times T_p = 4f\phi T_p$ <i>RMS value = form factor × average value</i> $= 1.11 \times 4f\phi T_p$ $E_{rms/phase} = 4.44f\phi T_p$	

Module 2			
	Q13 (a)	Explain the ASA method of determining the voltage regulation of alternator	7 marks
	Ans	The ASA (Ammeter-Slip-Ammeter) method is a simple and practical method used to determine the voltage regulation of an alternator. It involves measuring the stator current at two different load conditions, and using these measurements to calculate the voltage regulation of the alternator. The following are the steps involved in the ASA method: • Measure the open-circuit voltage (VOC) of the alternator using a voltmeter. • Connect an ammeter in series with one phase of the stator winding of the alternator, and load the alternator with a known resistive load. • Record the ammeter reading (I1) and the load voltage (V1) across the load. • Disconnect the load, and measure the slip of the alternator by applying a known DC voltage to the rotor winding and measuring the speed of the alternator using a tachometer. • Reconnect the load, and record the new ammeter reading (I2) and the new load voltage (V2) across the load. • Calculate the power factor (PF) of the alternator by dividing the measured load voltage by the measured line voltage, and taking the cosine of the result. • Calculate the apparent power (S) of the alternator using the formula $S = \sqrt{3} V_L I_L \text{ PF}$, where V_L is the line voltage and I_L is the average of the two stator currents (I1 and I2). • Calculate the real power (P) of the alternator using the formula $P = V_L I_L \text{ PF}$. • Calculate the voltage regulation (VR) of the alternator using the formula $\text{VR} = (\text{VOC} - V_2)/V_2 \times 100\%$. The ASA method is a simple and effective method of determining the voltage regulation of an alternator. It is widely used in the field for routine testing and maintenance of alternators.	
	Q13 (b)	A 1500 kVA, 6600 V, 3 phase, star connected alternator with a resistance of 0.4 ohm and reactance of 6 ohm per phase delivers full load current at power factor 0.8 lagging and normal rated voltage. Estimate the terminal voltage for the same excitation and load current at 0.8 p.f leading.	7 marks
	Ans	$V_L = 6600\text{V}$ $V_{ph} = 6600 / \sqrt{3} = 3810.512\text{V}$ $R_a = 0.4\Omega \quad X_s = 6\Omega$	

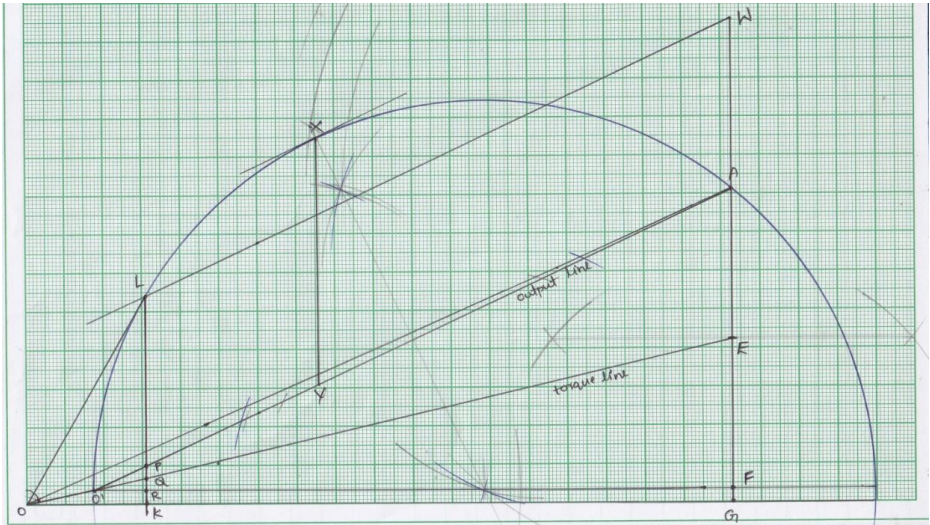
		Full load current, $I_a = \frac{P}{3V_{ph} \cos \phi} = \frac{1500 \times 10^3}{3 \times 3810.512 \times 0.8} = 164.02 A$ Induced EMF, $E_{ph} = \sqrt{(V_{ph} \cos \phi + I_a R_a)^2 + (V_{ph} \sin \phi + I_a X_s)^2}$ $= \sqrt{(3810.512 \times 0.8 + 164.02 \times 0.4)^2 + (3810.512 \times 0.6 + 164.02 \times 6)^2}$ = 4515.84 V	
	Q14 (a)	Explain the MMF method of determining the voltage regulation of alternator.	7 marks
	Ans	The MMF (magnetomotive force) method is a graphical method used to determine the voltage regulation of an alternator. It is based on the concept that the voltage regulation of an alternator is directly proportional to the ratio of the armature current and the field current. The following are the steps involved in the MMF method: • Measure the open-circuit voltage (VOC) of the alternator using a voltmeter. • Connect a resistive load to the alternator and adjust the field current to produce the rated voltage at the terminals of the alternator. • Measure the field current (If), armature current (Ia), and load voltage (VL) across the load. • Calculate the power factor (PF) of the alternator by dividing the measured load voltage by the measured line voltage, and taking the cosine of the result. • Calculate the apparent power (S) of the alternator using the formula $S = \sqrt{3} V_L I_a \text{ PF}$, where V_L is the line voltage and I_a is the armature current. • Calculate the field current required to produce rated voltage (If1) using the formula $I_{f1} = (VOC / \sqrt{3} V_L) \times 100\%$. • Calculate the armature reaction (Iar) using the formula $I_{ar} = \sqrt{I_a^2 - (S / V_L)^2}$. • Calculate the armature reaction ampere-turns (ATar) using the formula $AT_{ar} = 1.25 \times I_{ar} \times L$, where L is the length of the air gap in meters. • Calculate the field ampere-turns (ATf) required to produce rated voltage using the formula $AT_f = 1.25 \times I_{f1} \times L$, where L is the length of the air gap in meters. • Draw a graph with the y-axis representing armature reaction ampere-turns (ATar) and the x-axis representing field ampere-turns (ATf). • Plot the points (ATf, ATar) on the graph. • Draw a straight line through the plotted points. • Determine the slope of the line, which represents the voltage regulation of the alternator. The voltage regulation is given by: $VR = (VOC - V_L) / V_L \times 100\% = (\text{slope} / 1.25) \times 100\%$ The MMF method is a simple and effective method of determining the voltage regulation of an alternator. It is widely used in the field for routine testing and maintenance of alternators.	

Q14 (b)	A 10 kVA, 440 V, 50 Hz, 3 phase, star connected, alternator has the open circuit characteristics as below <table border="1"> <tr> <td>I_f (A)</td> <td>1.5</td> <td>3</td> <td>4</td> <td>5</td> <td>8</td> <td>11</td> <td>15</td> </tr> <tr> <td>V_{oc} (L-L)</td> <td>150</td> <td>300</td> <td>380</td> <td>440</td> <td>550</td> <td>600</td> <td>635</td> </tr> </table> With full load zero power factor, the excitation required is 14 amperes to produce 500 volts terminal voltage. On short circuit 4 amperes excitation is required to produce full load current. Determine the full load voltage regulation at 0.8 pf lagging by ZPF Method.	I_f (A)	1.5	3	4	5	8	11	15	V_{oc} (L-L)	150	300	380	440	550	600	635	7 marks
I_f (A)	1.5	3	4	5	8	11	15											
V_{oc} (L-L)	150	300	380	440	550	600	635											
Ans	To find full load current $P = \sqrt{3} V_L I_L$ $I_a = I_L = \frac{P}{\sqrt{3} V_L} = \frac{10000}{\sqrt{3} \times 440} = 13.122 A$  From Potier triangle, $RS = 2\text{cm}$, $PS = 3\text{cm}$ ie. $I_a X_L = 100 / \sqrt{3} = 57.735V$ $I_{f2} = 3A$ $E_1 = \sqrt{(V_{ph} \cos \phi + I_a R_a)^2 + (V_{ph} \sin \phi + I_a X_L)^2}$ $= \sqrt{(254.034 \times 0.8)^2 + (254.034 \times 0.6 + 57.735)^2} = 292.347V$ E_1 Line value = 506.36V $I_{f1} = 6.6 A$ $I_{f0} = \sqrt{I_{f1}^2 + I_{f2}^2 + 2 I_{f1} I_{f2} \cos(180 - (90 + \phi))}$ $\phi = \cos^{-1}(0.8) = 36.87^\circ$																	

		$I_{f0} = \sqrt{6.6^2 + 3^2 + 2 \times 6.6 \times 3 \times \cos(180 - (90 + 36.87))} = 8.736A$ $E_{line} = 568V$ $E_{ph} = 327.935V$ $\text{Voltage Regulation} = \frac{E_{ph} - V_{ph}}{V_{ph}} = \frac{327.935 - 254.034}{254.034} = 0.2909$ % Voltage Regulation = 29.09%	
Module 3			
	Q15 (a)	Sketch and explain V and inverted V curves of synchronous motor	7 marks
	Ans	(A) V curve of Synchronous Motor (B) Inverted V-curve of Synchronous Motor	
	Q15 (b)	Derive the expression for electromagnetic torque developed for induction motor	7 marks
	Ans	Torque produced in the induction motor depends on following factors: • Magnitude of rotating magnetic field • Magnitude of rotor current • Power factor of rotor circuit $T \propto \phi I_{2r} \cos(\phi_{2r})$ $\phi \propto E_1 \quad \left \quad I_{2r} = \frac{sE_2}{\sqrt{R_2^2 + (sX_2)^2}} \quad \right \quad \cos \phi_{2r} = \frac{R_2}{\sqrt{R_2^2 + (sX_2)^2}}$ $E_1 = \frac{E_2}{K}$ $T \propto E_2 \frac{sE_2}{\sqrt{R_2^2 + (sX_2)^2}} \frac{R_2}{\sqrt{R_2^2 + (sX_2)^2}}$ $T = \frac{ksE_2^2 R_2}{R_2^2 + (sX_2)^2}$ $T = \frac{3}{\omega_s} \frac{sE_2^2 R_2}{R_2^2 + (sX_2)^2}$ $k = \frac{3}{\omega_s}$ $\omega_s = 2\pi \frac{N_s}{60}$	

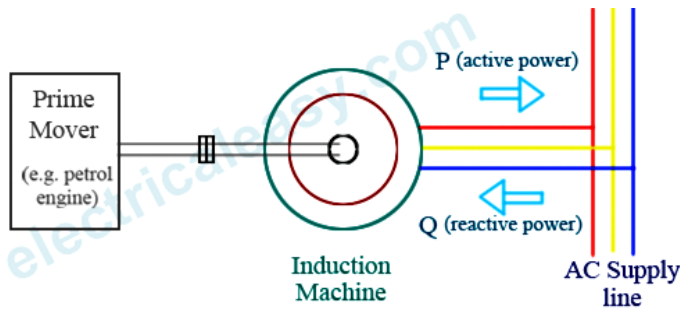
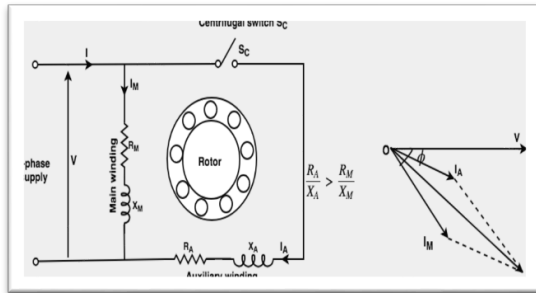
Q16 (a)	Explain the principle of operation of synchronous motor with neat sketch	7 marks
Ans	A synchronous motor is an AC motor that rotates at a constant speed determined by the frequency of the power supply and the number of poles in the motor. The synchronous motor operates based on the principle of magnetic fields being produced in the stator and rotor, which interact with each other to create a torque that drives the motor. The basic construction of a synchronous motor includes a stator, which contains the winding that produces a rotating magnetic field, and a rotor, which has a DC excitation source that produces a magnetic field. The rotor is made up of either permanent magnets or electromagnets that are excited by DC current. The principle of operation of a synchronous motor is based on the interaction of the stator magnetic field and the rotor magnetic field. When an AC voltage is applied to the stator winding, a rotating magnetic field is produced that rotates at a synchronous speed. This rotating magnetic field interacts with the magnetic field produced by the rotor, causing the rotor to rotate in synchronism with the stator magnetic field. The following are the steps involved in the operation of a synchronous motor: 1. When an AC voltage is applied to the stator winding, a rotating magnetic field is produced. The direction of the magnetic field rotates in a clockwise or counterclockwise direction based on the phase sequence of the applied voltage. 2. The rotating magnetic field produced by the stator winding induces a magnetic field in the rotor. 3. The magnetic field produced in the rotor interacts with the stator magnetic field, causing the rotor to start rotating in the same direction as the stator magnetic field. 4. The rotor will continue to rotate at a synchronous speed as long as the stator magnetic field is present. The synchronous motor can be used in a variety of applications, such as power generation, industrial drives, and railway traction systems, where precise speed control is required.  (a) Action of synchronous motor  (b) Action of synchronous motor	

Q16 (b)	The power input to the rotor of a 440v, 50 Hz, 6 pole, 3 phase induction motor is 80 kW. The rotor electromotive force is observed to make 100 complete alternations per minute. Calculate (a) the slip; (b) the rotor speed; (c) the mechanical power developed; (d) the rotor copper loss per phase; (e) the rotor resistance per phase if the rotor current is 65 A.	7 marks
Ans	a) The rotor electromotive force is observed to make 100 complete alternations per minute. Thus rotor frequency, $s f = 100/60 = 1.667$ Therefore, Slip $s = 1.667 / f = 1.667 / 50 = 0.033$ b) Synchronous speed, $N_s = 120f / P = 1000$ rpm Thus, Rotor speed, $N = N_s(1-s) = 966.77$ rpm c) Rotor input, $P_2 = 80\text{KW} = 80000$ W Mechanical power developed, $P_m = (1-s)P_2 = 77336$ W d) Rotor copper loss, $P_c = sP_2 = 2664$ W Rotor copper loss per phase = $2664 / 3 = 888$ W e) Rotor copper loss/phase = $I_2^2 R_2$ $888 = 65^2 * R_2$ Rotor resistance/phase, $R_2 = 0.2102\Omega$	
Module 4		
Q17 (a)	Justify the necessity of starters in three phase induction motors	4 marks
Ans	Starters are necessary for three-phase induction motors for various reasons, some of which are as follows: • To limit the inrush current: When a three-phase induction motor is started, a high inrush current flows through the motor windings, which can be several times the rated current of the motor. This inrush current can cause damage to the motor windings and other electrical components of the motor circuit. Starters are used to limit the inrush current by controlling the voltage and frequency applied to the motor during the starting process. • To control the starting torque: The starting torque of a three-phase induction motor is high, and it can cause damage to the driven equipment and motor itself. A starter can be used to control the starting torque of the motor by varying the voltage or current applied to the motor during the starting process. • To provide overload protection: Three-phase induction motors are susceptible to overload conditions, which can occur due to various reasons such as mechanical jamming of the driven equipment, improper cooling, and overvoltage or undervoltage conditions.	

	- Starters are designed to provide overload protection to the motor by sensing the current flowing through the motor and tripping the circuit breaker or contactor in case of overload conditions. - To provide reverse-phase protection: In a three-phase induction motor, if any two phases are interchanged, the motor will rotate in the opposite direction. Starters are designed to provide reverse-phase protection to prevent the motor from running in the wrong direction, which can cause damage to the driven equipment and motor. - Therefore, starters are essential for three-phase induction motors to ensure safe and reliable operation, protect the motor and driven equipment, and improve the efficiency of the motor.	
Q17 (b)	Draw the no load and short circuit diagram for a 20 h.p., 400 V, 50 Hz, 3 phase star-connected induction motor from the following data (line values): No load test : 400 V, 9A, $\cos\phi = 0.2$ Short circuit test : 200 V, 50 A, $\cos\phi = 0.4$. From the diagram find (a) the line current and power factor at full load, and (b) the maximum horse power.	10 marks
Ans	$\phi_o = \cos^{-1}(0.2) = 78.46^\circ$ $\phi_{sc} = \cos^{-1}(0.4) = 66.42^\circ$ Short circuit current at normal voltage, $I_{SN} = \frac{V_o}{V_{sc}} I_{sc}$ $= \frac{400}{200} \times 50 = 100 \text{ A}$ * Current scale 1 cm = 5 A 	

	$* W_{SN} = W_{sc} \left[\frac{V_o}{V_{sc}} \right]^2 = \sqrt{3} V_{sc} I_{sc} \cos \phi_{sc} \times \left(\frac{V_o}{V_{sc}} \right)^2$ $= \sqrt{3} \times 200 \times 50 \times 0.4 \left[\frac{400}{200} \right]^2$ $= 27712.813 \text{ W}$ $* \text{ Power scale division} = psd = \frac{W_{SN}}{ AG } = \underline{\underline{3379.61 \text{ W/cm}}}$ $* AW = \frac{P_{out}}{psd} \quad P_{out} = 20 \text{ HP} = 14920 \text{ W}$ $= \underline{\underline{4.4 \text{ cm}}}$ $* \text{ Length of OL} = 6.25 \text{ cm}$ $(a) \therefore \text{ Line current at full load} = 6.25 \times 5 \text{ A/cm}$ $= \underline{\underline{31.25 \text{ A}}}$ $\text{power factor} = \cos \phi = \frac{LK}{OL} = \frac{5.45}{6.25} = \underline{\underline{0.872}}$ $(b) \text{ Max. } P_{out} = xV \times psd = 6.45 \times 3379.61 = \underline{\underline{21798.48 \text{ W}}} = \underline{\underline{29.22 \text{ HP}}}$	
Q18 (a)	Explain dynamic braking of induction motors	4 marks
Ans	- Dynamic braking is a braking technique used to stop the rotation of an induction motor by converting the kinetic energy of the rotating motor into electrical energy, which is then dissipated as heat in a resistor or absorbed by the power grid. Dynamic braking is commonly used in applications where the motor needs to be stopped quickly or where the use of mechanical brakes is not feasible. - The basic principle of dynamic braking is to disconnect the motor from the AC power supply and then connect it to a braking resistor, which acts as a load on the motor. When the motor is disconnected from the power supply, it continues to rotate due to its inertia. As the motor rotates, it generates an electromotive force (EMF) that is proportional to its speed. This EMF is used to feed a braking resistor, which dissipates the electrical energy in the form of heat. - Dynamic braking is a useful technique for stopping induction motors quickly and efficiently. It is commonly used in applications such as cranes, elevators, and hoists, where rapid deceleration is required. However, it should be noted that dynamic braking generates a significant amount of heat, and the braking resistor must be sized appropriately to handle the heat dissipation.	
Q18 (b)	Determine approximately the starting torque of an induction motor in terms of full load torque when started by means of (a) star-delta switch; (b) an auto-transformer with 50% tapping. Ignore magnetizing current. The short circuit current of the motor at normal voltage is five times the full load current, and the full load slip is 5%.	10 marks

Ans	(a) Star-Delta Starting: When starting an induction motor with a star-delta switch, the motor is initially started in a low-voltage, high-current star configuration, and then switched to a high-voltage, low-current delta configuration once it has reached a certain speed. The starting torque of an induction motor using the star-delta starting method can be estimated as follows: Starting Torque (T_{start}) $\approx (1/3) \times (V_{delta}/V_{star})^2 \times T_{sfl}$ $T_{start} \approx (1/3) \times (1.732)^2 \times T_{sfl}$ $T_{start} \approx 1.5 \times T_{sfl}$ Therefore, the starting torque of an induction motor using the star-delta starting method is approximately 1.5 times the full load torque. (b) Auto-Transformer Starting: When starting an induction motor with an auto-transformer, the motor is initially started at a reduced voltage by tapping into the auto-transformer at a point lower than the full supply voltage. The starting torque of an induction motor using the auto-transformer starting method can be estimated as follows: Starting Torque (T_{start}) $\approx (V_{auto}/V_{full})^2 \times (T_{sfl}/2)$ Where: V_{auto} = the auto-transformer voltage V_{full} = the full supply voltage T_{sfl} = the full load torque Assuming that the auto-transformer has a 50% tapping, the voltage ratio (V_{auto}/V_{full}) would be 0.5, so we can simplify the equation as follows: $T_{start} \approx (0.5)^2 \times (T_{sfl}/2)$ $T_{start} \approx 0.125 \times T_{sfl}$ Therefore, the starting torque of an induction motor using the auto-transformer starting method is approximately 0.125 times the full load torque.	
Module 5		
Q19 (a)	Explain grid connected operation of induction generator.	6 marks
Ans	- Induction generator is not a self-excited machine. - Therefore, when running as a generator, the machine takes reactive power from the AC power line and supplies active power back into the line. - Reactive power is needed for producing rotating magnetic field. - The active power supplied back in the line is proportional to slip above the synchronous speed. Induction generators can be operated in two ways: - Grid connected operation - Self excited operation <u>Grid Connected Induction Generator</u> - An induction machine needs reactive power for excitation, regardless whether it is operating as a generator or a motor. - When an induction generator is connected to a grid, it takes reactive power from the grid.	

		
Q19 (b)	Explain the principle behind split phase method of starting of single phase induction motor	8 marks
Ans	◆ The split-phase induction motor is also known as a resistance-start motor. ◆ It consists of a single-cage rotor, and its stator has two windings, the main winding and a starting (also known as an auxiliary) winding. ◆ Both the windings are displaced by 90° in space like the windings in a two-phase induction motor. ◆ The main winding of the induction motor has very low resistance and high inductive reactance. 	
Q20 (a)	Explain double revolving field theory of single phase induction motor	8 marks
Ans	The double-revolving field theory is a simplified explanation of the operation of a single-phase induction motor. According to this theory, a single-phase induction motor operates by creating two revolving magnetic fields in the stator, which interact with a magnetic field produced by a single-phase AC supply to create a rotating magnetic field in the rotor. The stator of a single-phase induction motor contains a main winding and an auxiliary winding. The main winding is connected in series with the single-phase AC supply, while the auxiliary winding is connected in parallel with a capacitor. The capacitor helps to create a phase shift between the current and voltage in the auxiliary winding, which is necessary to create a second magnetic field in the stator. When the single-phase AC supply is applied to the main winding, a magnetic field is produced in the stator that rotates at the frequency of the AC supply.	

	The auxiliary winding, being connected in parallel with a capacitor, produces a second magnetic field that also rotates at the frequency of the AC supply, but in the opposite direction to the main magnetic field. The two magnetic fields produced by the main and auxiliary windings in the stator interact with the magnetic field produced by the single-phase AC supply, causing a rotating magnetic field to be produced in the rotor. The rotor is not energized, but it produces a magnetic field that is attracted to the rotating magnetic field in the stator, causing the rotor to rotate in the direction of the rotating magnetic field. The double-revolving field theory provides a simplified explanation of the operation of a single-phase induction motor. However, it does not provide a complete explanation of the motor's operation, as the actual motor operation is more complex and involves multiple magnetic fields and interactions between the stator and rotor. Nonetheless, the theory is useful for understanding the basic principles of single-phase induction motor operation.	
Q20 (b)	Explain the principle of operation of shaded pole motor	6 marks
Ans	A shaded pole motor is a type of single-phase induction motor that has a salient pole on the stator with a copper or aluminum shading coil or a short-circuited winding wound around a part of the pole's circumference. The shading coil provides a time-delayed magnetic field that produces a phase shift between the main winding and the shading coil, resulting in the production of a rotating magnetic field that drives the rotor. The operation of a shaded pole motor can be explained as follows: 1. When the motor is connected to a single-phase AC supply, a magnetic field is produced in the stator winding that rotates at the frequency of the AC supply. 2. The shading coil, which is positioned in close proximity to a portion of the pole face, creates a magnetic field that lags behind the main magnetic field produced by the stator winding. This time delay in the magnetic field produces a phase shift between the two magnetic fields. 3. The phase shift between the main winding and shading coil creates a rotating magnetic field in the stator. The rotating magnetic field is not uniform but has a lower flux density at the shaded part of the pole where the shading coil is wound. 4. The lower flux density causes the rotor to experience a greater magnetic reluctance, which results in a net force acting on the rotor in the direction of the unshaded part of the pole. 5. The net force causes the rotor to rotate in the direction of the unshaded part of the pole, and the motor starts to operate.	